J Therm Anal Calorim
DOI 10.1007/s10973-015-4757-z

Non-isothermal decomposition kinetics of $FeC_2O_4 \cdot 2H_2O$ prepared by solid-state method aiming at the formation of Fe_2O_3

Yu Feng[1] · Tianqi Hu[1] · Zhaoting Pu[1] · Mengmeng Wu[1] · Jie Mi[1]

Received: 20 December 2014 / Accepted: 3 May 2015

Abstract The thermal decomposition of $FeC_2O_4 \cdot 2H_2O$ prepared by solid-state method was investigated in air using TG-DTG/DSC. The precursor and its decomposition products formed at various temperatures were characterized by FT-IR and XRD. As evidenced by the results, the thermal decomposition proceeds in two steps with DSC peaks closely corresponding to the mass loss. The decomposition of $FeC_2O_4 \cdot 2H_2O$ and the kinetic analysis of the second step were performed under non-isothermal conditions, and the activation energies were estimated by the Ozawa integral method and the Coats–Redfern integral method. In order to make the kinetic calculation much more accurate, the temperature was risen by variable acceleration in the different stages of the decomposition to help distinguish the two steps from each other, and this is an innovation point that is deserved to be mentioned in this study. After calculation and comparison, the decomposition conforms to the nucleation and growth model and the physical interpretation is summarized. The activation energy and kinetic mechanism function were determined to be 113.90 kJ mol^{-1} and $G(\alpha) = [-\ln(1-\alpha)]^{2/3}$, respectively.

Keywords Thermal decomposition behavior · Iron oxalate · Ozawa integral method · Coats–Redfern integral method · Solid-state method · Nanostructured ferric oxide

✉ Jie Mi
mijie111@163.com

1 Key Laboratory of Coal Science and Technology of Ministry of Education and Shanxi Province, Taiyuan University of Technology, Taiyuan, People's Republic of China

Introduction

Sulfur compounds, mainly as H_2S and COS, are the most abundant impurity in coal-derived synthesis gas [1]. These species must be removed from the syngas prior to its utilization because of their negative effects on chemical processing and environment. Fe_2O_3 is known to be used as a high efficient desulfurizer in the medium–high temperature aiming at the coal-derived fuel gas and chemically synthesized gas cleanup [2, 3]. The iron oxides used as desulfurizer should always largely maintain the properties such as nanoscale, high dispersion and prodigious specific surface area, which greatly depend on the physical microstructure that largely influences the sulfur capacity of the desulfurizer. Therefore, the method of preparation is of great importance in obtaining the desulfurizer of optimum property.

Up to now, many methods such as sol–gel process, coprecipitation and microemulsion have been developed to synthesize nanometer Fe_2O_3 [4–6], while the thermal decomposition of $FeC_2O_4 \cdot 2H_2O$ aiming at obtaining Fe_2O_3 represents another typical group of preparation methods that have been intensively studied [7–13]. During the decomposition of metal oxalates, the release of carbon oxides plays a role of pore former, resulting in a substantial increase in pore structure along with large specific surface area and pore volume, which are beneficial to the dispersion of active component in the preparation of desulfurizer, and this is why the method was chosen in this study [14]. As for the preparation of $FeC_2O_4 \cdot 2H_2O$, compared with liquid-state reactions, the solid-state method has the advantages of flexibility of operation and simple technique such as no need of solvents, almost no pollution, high yields, low energy consumption and lower temperature [15, 16].

Broadbent et al. described the thermal decomposition of $FeC_2O_4{\cdot}2H_2O$ in nitrogen that was reflected by corresponding endothermic peak in the DTA curve, with free iron, Fe_3O_4 and FeO as the final decomposition products [9]. In the air, the final product turned out to be Fe_2O_3 only. And this has been verified by Zboril et al. and Koga et al. [8, 9, 17]. Even if the thermal decomposition of $FeC_2O_4{\cdot}2H_2O$ has been studied by many scholars, the mechanism and decomposition kinetic still remain defects, such as the overlapping between the thermal dehydration and the oxidative decomposition of anhydrous FeC_2O_4, which makes it difficult to distinguish the two decomposition steps clearly and calculate the kinetics accurately [18]. In order to solve this problem, the TG-DTG/DSC curves were recorded with temperatures risen by variable acceleration to distinguish the signals derived from the two peaks representing for the two steps, respectively, and after that, the problem has been successfully handled.

In this paper, $FeC_2O_4{\cdot}2H_2O$ was synthesized as a precursor by solid-state method. The thermal decomposition of $FeC_2O_4{\cdot}2H_2O$ and kinetic behaviors were monitored in air with TG-DTG/DSC, and techniques such as XRD and FT-IR were used to ensure the final products. Analysis methods including Ozawa integral method and Coats–Redfern integral method were applied to calculate the kinetic model and parameters of the decomposition [19–21]. The thermal analyses employed in the decomposition process can be significant to understand the transforms of intermediates, and this is good for choosing the appropriate temperature to prepare Fe_2O_3. Information gathered from the analyses and calculations helps gain more insights of the decomposition and can be directly applied in the design and preparation of Fe_2O_3-based desulfurizer.

Experimental

Preparation

$FeC_2O_4{\cdot}2H_2O$ was prepared by solid-state method via $FeCl_2{\cdot}4H_2O$ (analytical grade) and $H_2C_2O_4{\cdot}2H_2O$ (analytical grade) at room temperature. The original materials were mixed thoroughly at a molar ratio of 1.0 and homogenized in an agate mortar before oven-dried for 1 h. Then the powder mixture was levigated in a planetary ball crusher for 2 h. Finally, the mixture was dried in an oven at 80 °C for 2 h.

Characterization

The $FeC_2O_4{\cdot}2H_2O$ and intermediates during the heating process were characterized at several selected temperatures by X-ray diffraction (XRD) using a Rigaku D/max-2500 equipped with graphite monochromatized Cu Ka radiation source ($k = 1.5418$ Å) at the conditions of 40 kV and 100 mA.

Spectroscopic properties of $FeC_2O_4{\cdot}2H_2O$ and intermediates were measured by Fourier transform infrared (FT-IR) spectrometer via a SHIMADZU FTIR-8400s in the range of 4000–400 cm^{-1} at room temperature. The transmittance was plotted versus wave number (cm^{-1}) in the spectra.

TG-DTG/DSC measurements were performed in flowing air (50 mL min^{-1}) on a German NETZSCH-STA409C. A series of non-isothermal tests of $FeC_2O_4{\cdot}2H_2O$ were carried out from room temperature to 500 °C. In the instrument of TG-DTG/DSC, about 10.0 mg of the sample was placed in Pt crucibles and loosely covered and then heated at 5, 10, 15, 20 °C min^{-1} in the range of 25–120 and 190–500 °C, and heated at 1 °C min^{-1} from 120 to 190 °C in flowing air, respectively. All the experiments were carried out under identical conditions to assure the accuracy and uniformity of each measurement. The morphology and particle size of $FeC_2O_4{\cdot}2H_2O$ and its decomposition products were examined by scanning electronic microscopy (ZEISS, EVO-MA15).

Results and discussion

Thermal analysis

TG-DTG/DSC curves of $FeC_2O_4{\cdot}2H_2O$ are represented in Fig. 1. A small but distinct inflexion point and a plateau exist at 180 °C, indicating that the decomposition proceeded in two separated steps, the dehydration and the oxidation decomposition. The first step is in the range of 120–180 °C, documented by an endothermic peak in the DSC curve with a maximum at 172 °C with a 20.97 mass%

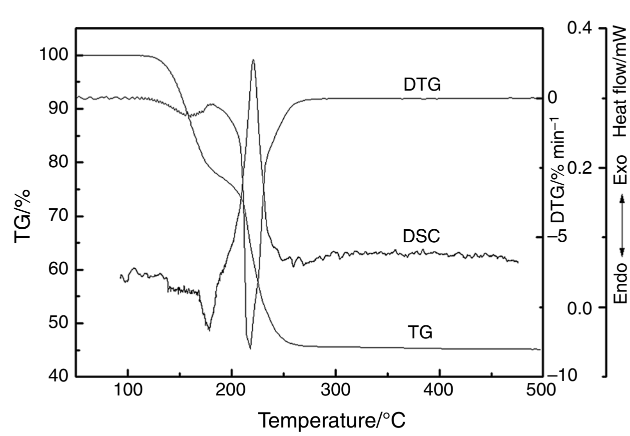

Fig. 1 TG-DTG/DSC curves of $FeC_2O_4{\cdot}2H_2O$ calcined at a heating rate of 5 °C min^{-1}

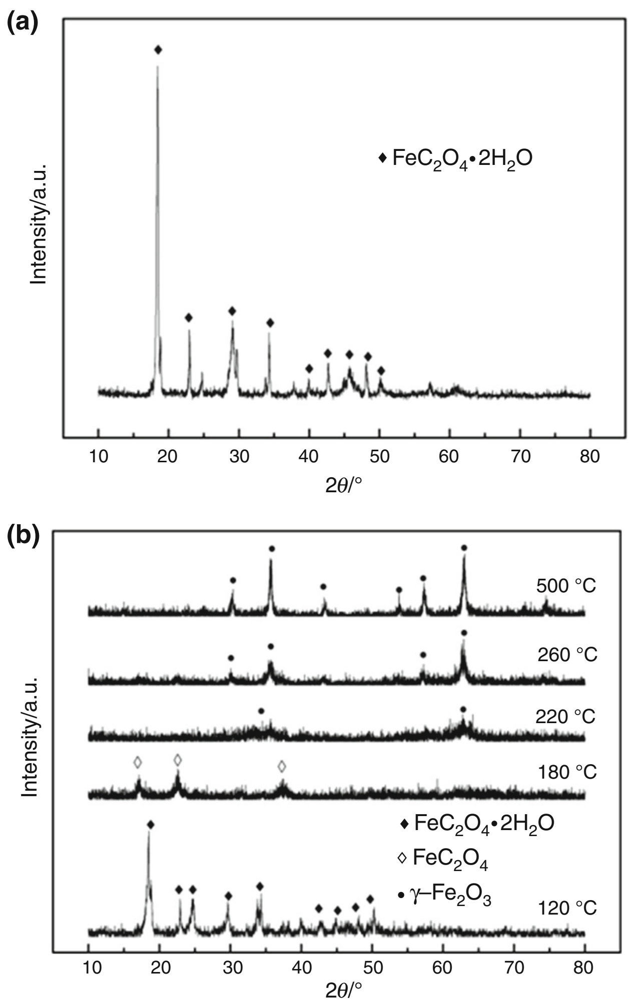

Fig. 2 XRD patterns of $FeC_2O_4 \cdot 2H_2O$ (**a**) and the products calcined at different temperatures (**b**)

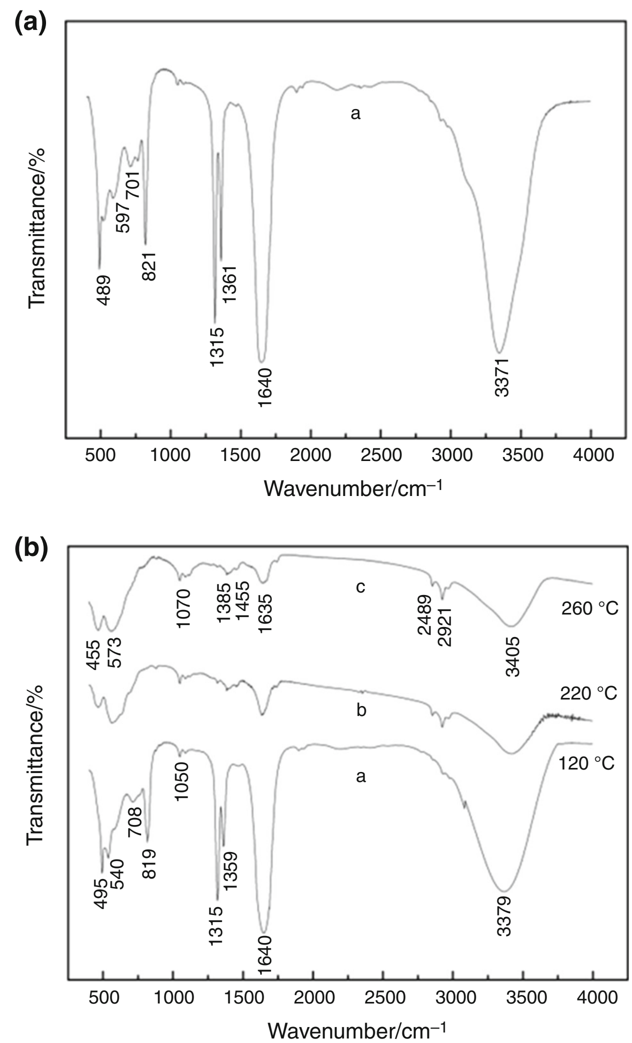

Fig. 3 FT-IR spectra of $FeC_2O_4 \cdot 2H_2O$ (**a**) and the products prepared at different temperatures (**b**)

mass loss that is in accordance with the expectation of the evolution of two molecules of H_2O [22, 23]. The second step is the decomposition of anhydrous FeC_2O_4, which is represented by a sharp exothermic peak in the DSC curve at 217 °C. The decomposition accounts for a total mass loss of 55.0 mass% that is corresponding to the formation of γ-Fe_2O_3, which is a common fact among the thermal decomposition in air [22, 23]. This is further confirmed by the XRD patterns derived from the final conduct of the decomposition. While, at 500 °C, the terminal decomposition temperature of the second step, no further phase transition that was recorded in precious studies was found here from the DSC curves and XRD patterns. Therefore, the corresponding chemical equations can be described as [8, 9]:

$$FeC_2O_4 \cdot 2H_2O \rightarrow FeC_2O_4 + 2H_2O \quad (1)$$

$$FeC_2O_4 + O_2 \rightarrow Fe_2O_3 + CO_2 \quad (2)$$

X-ray powder diffraction

The characteristic XRD patterns are represented in Fig. 2. In Fig. 2a, the XRD patterns are in consistent with the new peaks originated from solid-state method, which are assigned to the diffractions of $FeC_2O_4 \cdot 2H_2O$ [8, 9], while no characteristic peaks of $FeCl_2 \cdot 4H_2O$ and $H_2C_2O_4 \cdot 2H_2O$ can be seen here anymore. As observed in Fig. 2b, when the decomposition temperature rises to 120 °C, there is no significant difference in the XRD patterns compared with that of precursor but just shows low-intensity lines due to partial dehydration of the precursor. With the increase in temperature, the peak intensity coincides with the $FeC_2O_4 \cdot 2H_2O$ decreases, while that of FeC_2O_4 appears and increases at 180 °C [8, 9]. The diffraction lines of γ-Fe_2O_3 show up as the temperature rises to 220 °C [8, 24]. A much

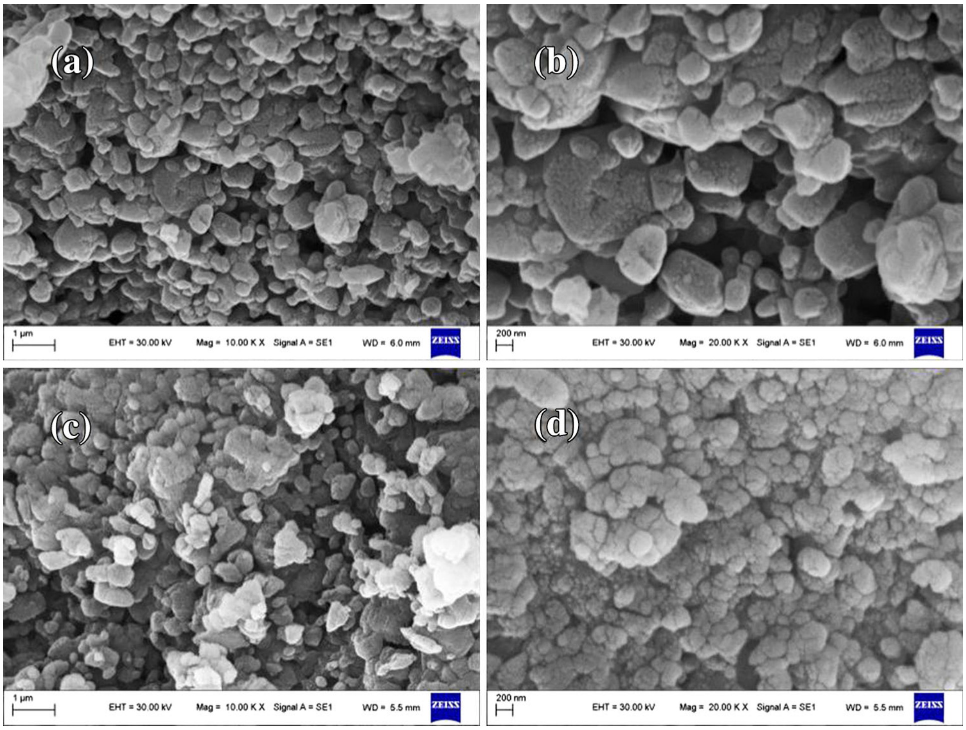

Fig. 4 SEM graphs of $FeC_2O_4{\cdot}2H_2O$ (**a**, **b**) and γ-Fe_2O_3 (**c**, **d**)

sharper peak assigned to γ-Fe_2O_3 along with a higher crystallinity has been noted with the increase in temperature.

Vibrational spectroscopy

The FT-IR spectra of $FeC_2O_4{\cdot}2H_2O$ and the solid products are shown in Fig. 3. The presence of water that exists in the structure of oxalate was revealed by the characteristic bands at 3371 cm^{-1}, identified as O–H stretching and bending modes of vibration. The strong band at 1640 cm^{-1} is assigned to asymmetric C–O, and the closely spaced bands at 1361 and 1315 cm^{-1} are characteristic of symmetric O–C–O stretching modes [25]. The characteristic bending modes for C = C–O and O–C–O are observed at 821 and 489 cm^{-1}, respectively, which show the typical absorption bands of metal oxalate hydrates [26]. From all the above, the parent sample is $FeC_2O_4{\cdot}2H_2O$ without any other impurity [9, 27]. By raising the calcination temperature up, the attenuation and disappearance of the characteristic bands indicate the complete decomposition of the oxalate. Bands located at 455 and 573 cm^{-1} are the typical absorption bands belonged to the characteristic bands of the stretching vibrations of Fe–O [28]. The bands of O–H that still exist at 3371 cm^{-1} are due to the water molecules that were absorbed by nanometer granules of γ-Fe_2O_3. The band persists even upon raising the temperature up to 500 °C, which was mentioned by Gabal [29, 30].

Scanning electron microscopy

The particle morphologies and textures of the precursor and the final product are shown in Fig. 4. As observed in Fig. 4a, b, the particle of $FeC_2O_4{\cdot}2H_2O$ is mainly in a shape of irregular ellipse with its size varying in the range of 200–500 nm, and the particle size of γ-Fe_2O_3 is about 50–100 nm as observed in Fig. 4c, d. While the slight agglomerates caused by moisture absorbed from air are also seen. However, what should be noted is that because of the relatively poor technique conditions, much more smaller spherical and elliptical pellets in size ranged from 20 to 30 nm are visible for naked eyes, but cannot be distinguished by instrument clearly.

Thermal decomposition kinetic of FeC_2O_4

The DSC curves derived from the decomposition process are shown in Fig. 5. As shown in Fig. 5, the peak temperature of the thermal decomposition shifts to the higher temperature with the increase in heating rate and the peaks become more and more sharp. While during the dehydration, the heating rate is 1 °C min^{-1} only, so the dehydration temperature is nearly the same. Kinetic analysis under non-isothermal measurements for the anhydrous FeC_2O_4 decomposition was carried out using different heating rates, assuming different solid-state reaction equations in view of two integral methods: Ozawa method and Coats–

Redfern method. In Ozawa method, the relevant data come from the thermogravimetry curves [18, 29]. The decomposition proceeded in accordance with the Ozawa method is approximated to the formula below:

$$\lg \beta = \lg[AE/RG(\alpha)] - 2.315 - 0.4567E/RT \quad (3)$$

where $\alpha(0 < \alpha < 1)$, A, E, R, T are the fractional conversion rate, pre-exponential factor, activation energy, universal gas constant and absolute temperature. With dynamic techniques, temperature rate dT/dt is set to be a constant β. $G(\alpha)$ stands for the conversion functional relationship. Different forms of $G(\alpha)$ referenced in this paper are based on the literature [30]. The heating rate and the degree of conversion are defined as follows, respectively:

$$\beta = \mathrm{d}T/\mathrm{d}t \quad (4)$$

$$\alpha = (m_0 - m)/(m_0 - m_f) \quad (5)$$

The m_0, m and m_f refer to the initial, actual and final mass of the sample.

The kinetic equation was obtained by integrating the Arrhenius equation after assuming that the kinetic model and the kinetic parameters were invariant all over the process, so it is obvious that the calculation of E_β is independent of the explicit form of $G(\alpha)$ and reaction model that used to describe the reaction.

The reaction rate at a constant extent of conversion is the only function of the temperature [31, 32], and the equation is written as follows:

$$\lg \beta = C - 0.4567E/RT \quad (6)$$

Measure the right transformational temperature at the same conversion rate with the different heating rates. Then the ln β versus $1/T$ plot must give a straight line. After a linear least square fit, the slope of the line is just the

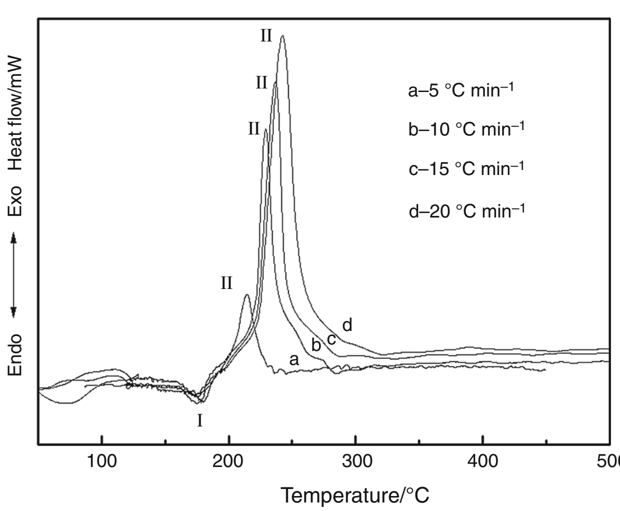

Fig. 5 DSC curves of the thermal decomposition of $FeC_2O_4{\cdot}2H_2O$ under different heating rates

Table 1 Basic data needed for the Ozawa method

β/K min^{-1}	lg β	T_p/K	$1/T_p$
5	0.699	490.56	0.002038
10	1.000	502.33	0.001990
15	1.176	509.78	0.001961
20	1.301	514.76	0.001942

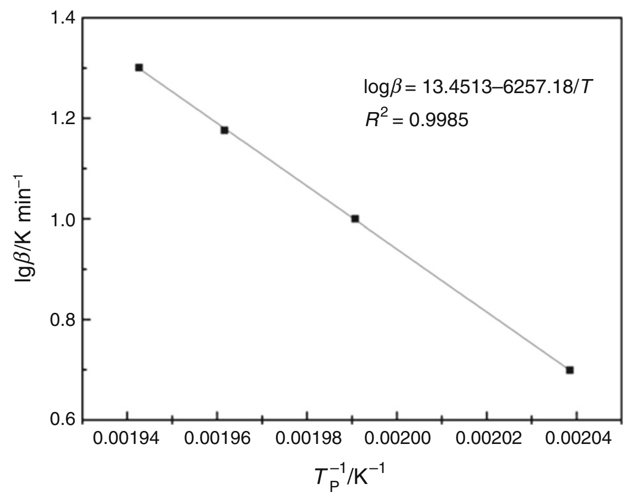

Fig. 6 Plot of lg β versus $1/T_P$ for the decomposition of FeC_2O_4

activation energy E_β. The temperatures corresponding to the conversion rate α ($0.1 \leq \alpha \leq 0.8$) are selected as the basic data. The data calculated are shown in Table 1. A straight line alone with a correlation coefficient $R^2 = 0.9985$ is traced in Fig. 6, in which the logarithm of heating rates is plotted against the reciprocal of peak absolute temperatures. The activation energy E_β that can be obtained from the slope of the line turned out to be 113.90 kJ mol^{-1}, and the linear equation can be described as follows:

$$\lg \beta = 13.4513 - 6257.18/T \quad (7)$$

The Coats–Redfern method is described with the following equation [32]:

$$\ln[G(\alpha)/T^2] = \ln(AR/\beta E) - E/RT \quad (8)$$

The terms in Eq. (8) have the same meaning with those in Eq. (3). The results obtained at a constant heating rate with different α values were superimposed on one master curve, from which the basic data needed for Coats–Redfern method are summarized in Table 2. The plot of $\ln[G(\alpha)/T^2]$ versus $1/T$ is shown in Fig. 7. The activation energy E_α can be calculated from the linear equation that was derived from the plot of $\ln[G(\alpha)/T^2]$ versus $1/T$. In general, if an equation exists as $E_\alpha \approx E_\beta$, and the E_α and E_β were calculated from Ozawa method and Coats–Redfern method,

Table 2 Absolute temperatures corresponding to different α under the heating rate 5 °C min^{-1}

α	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8
T/K	475.66	483.78	488.22	491.87	496.02	498.02	502.08	505.47

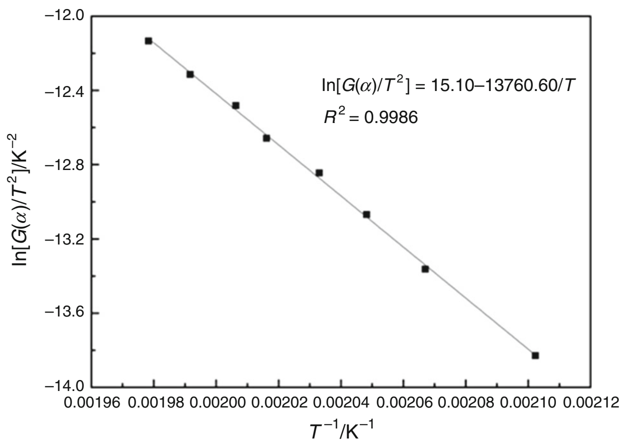

Fig. 7 Plot of ln[$G(\alpha)/T^2$] versus $1/T$ of the decomposition of FeC_2O_4

respectively, namely the activation energy E is exactly the activation energy of the process [30, 32]. After careful calculation, the most similar E_β obtained using Coats–Redfern method by comparing with E_α is 114.41 kJ mol^{-1}, which has the lowest deviation of 0.5 kJ mol^{-1} due to the A_n ($n = 3/2$) function, namely an error of 0.45 %. The function $G(\alpha)$ corresponding to E_α is regarded as the optimum kinetic conversion function, which is written in the form:

$$G(\alpha) = [-\ln(1-\alpha)]^{2/3} \tag{9}$$

The linear equation can be shown as:

$$\ln[G(\alpha)/T^2] = 15.10 - 13760.60/T \tag{10}$$

The results show that the optimum value of activation energy E is 113.90 kJ mol^{-1}, and the maximum possible conversion function obtained from the second step decomposition of $FeC_2O_4{\cdot}2H_2O$ is the Avrami–Erofeev random nucleation, namely the A_n ($n = 3/2$) model. What should be mentioned here is that every calculation and measurement has been carefully calculated twice or even more to ensure the accuracy and repeatability.

Conclusions

In summary, the $FeC_2O_4{\cdot}2H_2O$ can be efficiently synthesized by solid-state method with $FeCl_2{\cdot}4H_2O$ and $H_2C_2O_4{\cdot}2H_2O$ as raw materials. The formation of γ-Fe_2O_3 which is the final product of the decomposition was confirmed in a two-step decomposition. A detailed mechanism of the process has been proposed through analyses and calculations. The activation energy E is 113.90 kJ mol^{-1} ensured by Ozawa integral method and Coats-Redfern method. The decomposition conformed to the model of random nucleation and subsequent growth, corresponding to the A_n ($n = 3/2$) model, and the decomposition conversion function can be described as $G(\alpha) = [-\ln(1-\alpha)]^{2/3}$. And surely the research of the decomposition helps ensure the proper calcination temperature to control the particle size and the purity of the metal oxides.

Acknowledgements The authors gratefully acknowledge that this work is supported by the National Natural Science Foundation of China (51272170/21276172).

References

1. Dolan MD, Ilyushechkin AY, McLennan KG, Nguyen T, Sharma SD. Glass-based processing of mixed-oxide desulfurization sorbents. Ind Eng Chem Res. 2009;48:10498–503.
2. Slimane RB, Abbasian J. Regenerable mixed metal oxide sorbents for coal gas desulfurization at moderate temperatures. Adv Environ Res. 2000;4:147–62.
3. Zhao R, Shangguan J, Lou Y, Song J, Mi J, Fan H. Regeneration of Fe_2O_3-based high-temperature coal gas desulfurization sorbent in atmosphere with sulfur dioxide. Front Chem Eng Chin. 2010;4:423–8.
4. Ennas G, Marongiu G, Musinu A, Falqui A, Ballirano P, Caminiti R. Characterization of nanocrystalline γ–Fe_2O_3 prepared by wet chemical method. J Mater Res. 1999;14:1570–5.
5. Jeong JR, Lee SJ, Kim JD, Shin SC. Magnetic properties of γ-Fe_2O_3 nanoparticles made by coprecipitation method. Phys Status Solidi B. 2004;241:1593–6.
6. Nakanishi T, Iida H, Osaka T. Preparation of iron oxide nanoparticles via successive reduction–oxidation in reverse micelles. Chem Lett. 2003;32:1166–7.
7. Broadbent D, Dollimore D, Dollimore J. The thermal decomposition of oxalates. Part IX. The thermal decomposition of the oxalate complexes of iron. J Chem Soc A Inorg Phys Theor. 1967; 451–4.
8. Zboril R, Machala L, Mashlan M, Hermanek M. Nanocrystalline iron (III) oxides formed under dynamic heating of ferrous oxalate dihydrate in air. In: Industrial applications of the Mossbauer effect: international symposium on the industrial applications of the Mossbauer effect. AIP Publishing; 2005. p. 257–62.
9. Koga N, Sato Y. Formation and transformation kinetics of amorphous iron (III) oxide during the thermally induced transformation of ferrous oxalate dihydrate in air. J Phys Chem A. 2010;115:141–51.
10. Mohamed MA, Galwey AK. A kinetic and mechanistic study of the isothermal decomposition of ferrous oxalate dihydrate. Thermochim Acta. 1993;213:269–78.

11. Cetişli H, Çılgı GK, Donat R. Thermal and kinetic analysis of uranium salts. J Therm Anal Calorim. 2012;108:1213–22.
12. Çılgı GK, Cetişli H, Donat R. Thermal and kinetic analysis of uranium salts. J Therm Anal Calorim. 2014;115:2007–20.
13. Altaş Y, Eral M, Tel H. Preparation of homogeneous $(Th_{0.8\text{-}}U_{0.2})O_2$ powders by mechanical blending of $Th(C_2O_4)_2{\cdot}6H_2O$ and $U(C_2O_4)_2{\cdot}6H_2O$ powders. J Nucl Mater. 2001;294:344–8.
14. Efthimiadis EA, Sotirchos SV. Effects of pore structure on the performance of coal gas desulfurization sorbents. Chem Eng Sci. 1993;48:1971–84.
15. Wu J, Yuan AQ, Huang ZY, Tong ZF, Chen J, Liang RL. Thermochemical properties and decomposition kinetics of ammonium magnesium phosphate monohydrate. Chin J Chem. 2007;25:72–6.
16. Liu H, Li Y, Xiao S, et al. Synthesis of organic one-dimensional nanomaterials by solid-phase reaction. J Am Chem Soc. 2003;125:10794–5.
17. Tang WJ, Chen DH. Thermal decomposition kinetics of ferrous oxalate dehydrate. Acta Phys Chim Sin. 2007;23:605–8.
18. Pan YX, Guan XY, et al. Study on the kinetic mechanism of the dehydration process of $FeC_2O_4{\cdot}2H_2O$ using double extrapolation. Acta Phys Chim Sin. 1998;12:1088–93.
19. Ozawa T. A new method of analyzing thermogravimetric data. Bull Chem Soc Jpn. 1965;38:1881–6.
20. Ozawa T. Kinetic analysis of derivative curves in thermal analysis. J Therm Anal. 1970;2:301–24.
21. Coats AW, Redfern JP. Kinetic parameters from thermogravimetric data. Nature. 1964;201:68–9.
22. Dollimore D, Griffiths DL, Nicholson D. The thermal decomposition of oxalates. Part II. Thermogravimetric analysis of various oxalates in air and in nitrogen. J Chem Soc. 1963;488:2617–23.
23. Macklen ED. Influence of atmosphere on the thermal decomposition of ferrous oxalate dehydrate. J Inorg Nucl Chem. 1967;29:1229–34.
24. Sreeja V, Joy PA. Microwave hydrothermal synthesis of γ-Fe_2O_3 nanoparticles and their magnetic properties. Mater Res Bull. 2007;42:1570–6.
25. D'Antonio MC, Wladimirsky A, Palacios D, Coggiolaa L, González-Baró AC, Baran EJ, Mercader RC. Spectroscopic investigations of iron (II) and iron (III) oxalates. J Braz Chem Soc. 2009;20:445–50.
26. Angermann A, Töpfer J. Synthesis of nanocrystalline Mn–Zn ferrite powders through thermolysis of mixed oxalates. Ceram Int. 2011;37:995–1002.
27. Gabal MA, Ata-Allah SS. Concerning the cation distribution in $MnFe_2O_4$ synthesized through the thermal decomposition of oxalates. J Phys Chem Solids. 2004;65:995–1003.
28. Morales MP, Veintemillas-Verdaguer S, Montero MI, Serna CJ, Roig A, Casas L, Sandiumenge F. Surface and internal spin canting in γ-Fe_2O_3 nanoparticles. Chem Mater. 1999;11:3058–64.
29. Salla JM, Morancho JM, Cadenato A, et al. Non-isothermal degradation of a thermoset powder coating in inert and oxidant atmospheres. J Therm Anal Calorim. 2003;72:719–28.
30. Vlaev L, Nedelchev N, Gyurova K, Zagorcheva M. A comparative study of non-isothermal kinetics of decomposition of calcium oxalate monohydrate. J Anal Appl Pyrol. 2008;81:253–62.
31. Sergey V. Modification of the integral isoconversional method to account for variation in the activation energy. J Comput Chem. 2001;22:178–83.
32. Hu C, Mi J, Shang S, et al. The study of thermal decomposition kinetics of zinc oxide formation from zinc oxalate dehydrate. J Therm Anal Calorim. 2014;115:1119–25.